

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · COMBINED SCIENCE

COMBINED SCIENCE 5006/1

PAPER 1 November 2019

1 hour

PAPER 1 — MULTIPLE CHOICE — 40 QUESTIONS — 1 HOUR

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2019 5006/1.

Additional materials: Answer all 40 questions. For each question there are four possible answers A, B, C and D.

INSTRUCTIONS: Choose the one you consider correct and record it. Each question carries 1 mark.

1. Diffusion is fastest in a [1]

- A) gas
- B) liquid
- C) crystal lattice at 0 K
- D) solid

2. Convection occurs in [1]

- A) a vacuum
- B) solids only
- C) fluids (liquids and gases)
- D) crystals only

3. A unit of density is [1]

- A) J
- B) N
- C) g/cm³
- D) W

4. Salivary amylase begins the digestion of [1]

- A) starch to maltose
- B) fat to glycerol
- C) protein to amino acids
- D) fibre to glucose

5. Oxygenated blood is pumped to the body from the [1]

- A) pulmonary vein only, without the heart
- B) right ventricle
- C) right atrium
- D) left ventricle

6. The energy released in respiration is used in the body to [1]

- A) stop diffusion
- B) create new elements
- C) drive processes such as muscle contraction and active transport
- D) make chlorophyll in blood

7. Distilled water, compared with tap water, contains [1]

- A) more dissolved salts
- B) fewer dissolved salts
- C) more chlorine always
- D) sand

8. A herbivore is a [1]

- A) primary consumer
- B) decomposer only
- C) pathogen
- D) producer

9. The word equation for photosynthesis is [1]

- A) carbon dioxide + water → glucose + oxygen
- B) glucose + oxygen → carbon dioxide + water
- C) glucose + carbon dioxide → oxygen + water
- D) nitrogen + water → glucose + oxygen

10. Which statement is correct? [1]

- A) Capillaries have thick muscular walls
- B) Arteries carry blood away from the heart
- C) Xylem carries blood
- D) Veins have the highest blood pressure

11. The SI unit of speed is [1]

- A) m/s
- B) J
- C) kg
- D) N

12. A gas has [1]

- A) fixed volume, no fixed shape
- B) no fixed shape or volume
- C) fixed shape and volume
- D) particles that cannot move

13. Ciliated cells in the trachea [1]

- A) pump blood
- B) sweep mucus and trapped particles
- C) absorb digested food
- D) make insulin

14. Sodium chloride is held together by [1]

- A) covalent bonds only
- B) metallic bonds only
- C) hydrogen bonds only
- D) ionic bonds

15. The pitch of a sound depends on its [1]

- A) colour
- B) frequency
- C) amplitude
- D) loudness only

16. The main function of red blood cells is to **[1]**

- A) clot blood
- B) make urea
- C) transport oxygen
- D) fight all pathogens by phagocytosis only

17. In human muscle during vigorous exercise, anaerobic respiration produces **[1]**

- A) ethanol only
- B) lactic acid
- C) carbon dioxide and water
- D) oxygen

18. The function of the nucleus is to **[1]**

- A) store starch only
- B) be a fluid-filled sac only
- C) control cell activities
- D) make glucose

19. Isotopes of an element have **[1]**

- A) the same proton number but different nucleon number
- B) the same mass number but different proton number
- C) no neutrons
- D) different numbers of protons always

20. Electrolysis of aqueous copper(II) sulfate with copper electrodes is used to **[1]**

- A) purify copper
- B) make sodium
- C) liquefy air
- D) extract aluminium

21. The kidney **[1]**

- A) makes red cells
- B) produces bile
- C) removes urea and excess water from the blood
- D) stores glycogen only

22. A plant cell in a concentrated sugar solution may become **[1]**

- A) photosynthetic
- B) turgid
- C) plasmolysed
- D) a gamete

23. A dull black surface, compared with a shiny silver surface, **[1]**

- A) is a poor absorber of radiation
- B) is a good absorber and emitter of radiation
- C) cannot emit radiation
- D) reflects all heat always

24. Plasma transports **[1]**

- A) dissolved substances such as carbon dioxide, urea and hormones
- B) only starch grains
- C) only oxygen in haemoglobin
- D) only cellulose

25. Blue litmus paper in an acid [1]

- A) stays blue
- B) turns green
- C) turns red
- D) bleaches white always

26. Which is a correct order in treating river water for drinking? [1]

- A) boiling → mining
- B) electrolysis → rusting
- C) screening → filtration → chlorination
- D) chlorination → screening

27. A unit of density is [1]

- A) J
- B) W
- C) g/cm^3
- D) N

28. The rate of a reaction increases when [1]

- A) concentration decreases
- B) the surface area of a solid reactant increases
- C) the catalyst is removed
- D) temperature decreases

29. Magnetic field lines around a bar magnet [1]

- A) do not exist in air
- B) are always circular like a ripple only
- C) start at south and end at north
- D) start at north and end at south, outside the magnet

30. Red blood cells are adapted to carry oxygen because they contain [1]

- A) haemoglobin
- B) chlorophyll
- C) cellulose
- D) starch

31. The electronic configuration of chlorine (proton number 17) is [1]

- A) 2,8,7
- B) 2,8,2
- C) 2,8,1
- D) 2,6

32. Neutral solutions have pH [1]

- A) 7
- B) 14
- C) 1
- D) 0

33. Sound cannot travel through [1]

- A) air
- B) a vacuum
- C) steel
- D) water

34. Ionic bonding is the **[1]**

- A) sharing of a pair of electrons
- B) hydrogen bonding in water only
- C) attraction between nuclei only in metals
- D) transfer of electrons from metal to non-metal, forming ions

35. The SI unit of force is the **[1]**

- A) kilogram
- B) joule
- C) newton
- D) watt

36. The SI unit of mass is the **[1]**

- A) newton
- B) metre per second
- C) degree Celsius
- D) kilogram

37. For main-group elements, the group number indicates the **[1]**

- A) mass number
- B) number of electron shells
- C) number of outer-shell electrons
- D) number of neutrons

38. Pressure is defined as **[1]**

- A) $F + A$
- B) $F \times A$
- C) A / F
- D) F / A

39. A metal kettle is safer if it has **[1]**

- A) no insulation
- B) a higher current always
- C) a thicker live wire only
- D) an earth wire and a fuse / circuit breaker

40. Which statement is correct? **[1]**

- A) Conduction occurs only in gases
- B) Convection occurs in solids only
- C) Radiation does not need a medium
- D) Conduction is bulk movement of a fluid

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).